

Tenure hero pipeline — MBB mirror (use existing panel first)

Last synced: 2026-09-01

Audience: Charles + COMPASS (Mac analysis).

Standalone: Step-by-step map from rsync'd `faculty_panel_with_pools.jsonl` to tenure HERO plots, mirroring MBB post-re-entry work without re-scraping.

MBB analogue doc: `../MBB_empirical_roster_select_replay.md`

Side-by-side ladder

Rung	MBB (done or in progress)	Tenure (this campaign)
Data spine	MBB panel CSV / exports	<code>faculty_panel_with_pools.jsonl</code>
Pass A HERO	<code>pass_a_empirical_bundle.py</code> → <code>sports_sandbox/hero/</code>	<code>tenure_pass_a_hero.py</code> → <code>tenure_sandbox/hero/</code>
Pass A F-HERO	<code>pass_a_congestion_conditional.py</code> → <code>fhero/</code>	Act II — T_j at fixed $\hat{A}$ (pending lock)
ρ / H_{sort}	Reigning hero calibration; $p^* \approx 0 \rightarrow$ park for MBB	Act III — run diagnostic; park only if flat
Generative replay	<code>reigning_hero_sim_hero.py</code>	Act IV — after Alex lock
Scrape / expand	Box QC, min20, mg10	Parallel — PEER on Rivanna when scheduled

Step 0 — Confirm Mac has panel (once per session)

```
./scripts/rsync_pull_recent_hpc.sh tenure # if unsure
ls -lh tenure/tenure_pipeline/faculty_panel_with_pools.jsonl
```

See `20260901_PEER_to_COMPASS_tenure_hero_mac_handoff.md`.

Step 1 — Inference filter (strict v0)

From person-year rows, keep rows where:

- `match_confidence` $\in \{\text{HIGH, MEDIUM}\}$
- `poolq_loo_mean` is not null (on assistant rows used for LOO)
- `ever_assistant` is true

Do not hero-plot raw 106K without understanding OpenAlex NONE tier (62K person-years).

Step 2 — Collapse to person-level (HERO grain)

Same logic as `tenure/tenure_pipeline/stage9_analysis.py`:

- One record per `faculty_id`
- `loo_mean` = mean of `poolq_loo_mean` over **assistant** years with non-null LOO
- Drop persons with no computable LOO assistant years
- Person flags: `tenure_event`, `attrition`, `censored`

Step 3 — Outcomes (Option A — locked for v0)

Resolved = tenure OR attrition (known ending).

Rate	Numerator	Denominator
Tenure rate	<code>tenure_event</code>	resolved in bin
Attrition rate	<code>attrition</code>	resolved in bin

Censored: report `n_censored` per bin; **exclude** from rate denominator. Survival / Cox later.

Step 4 — Bin and plot

Knob	v0
X	Person-level <code>loo_mean</code>
Method	Quantile (equal people per bin)
Count	16
CI	Wilson on resolved counts
Outputs	PNG + CSV + provenance JSON → <code>tenure_sandbox/hero/</code>

Stage 9 (`stage9_inverted_u.png` , Apr 2026) used 18 bins on a **broader** population — useful sanity check, **not** the v0 deliverable.

Step 5 — Act II+ (later)

Act	Action
II F-HERO	Bin department mean pubs at fixed $\hat{A}$ slice
III ρ	Department-level H_{sort} bracket (compare to MBB reigning hero ρ folder)
IV Sim	Frozen departments → fit score knobs → sim tenure flags → replay HERO

ρ policy: Do **not** assume $\rho \approx 0$ on tenure. Run Act III before parking ASSIGN like MBB.

Code map

File	Role
<code>tenure/tenure_pipeline/stage9_analysis.py</code>	Reference implementation (collapse + bin + plot)
<code>tenure/tenure_pipeline/build_faculty_panel_inference_v1.py</code>	Inference tier filter reference
<code>tenure/scripts/tenure_pass_a_hero.py</code>	v0 CLI → <code>tenure_sandbox/hero/</code>
<code>sports/scripts/pass_a_empirical_bundle.py</code>	MBB template for provenance + filenames

Thread log

Date	Entry
2026-09-01	Initial pipeline map; v0 locks from Charles + Alex direction.